# User Acceptance Testing Plan for Network Traffic Profiler Dashboard

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# Scope

## Objectives and business requirements

The goal of this user acceptance test is to find out whether the project's target user base of non-technical experts can intuitively and properly use our PCAP file analysis system, with our AI model correctly classifying the contained user actions within the file.

We're planning to finish our planned part of the development and leave the project to be continued and improved by other developers, with the future objective of integrating the system into a network firewall.

We will measure success by evaluating PCAP upload times, network feature extraction, ease of readability, and the accuracy of the AI classification, using different PCAP files for the input.

## Scope (Tomiris)

Our development phase has been nearing completion, and we need to ensure that our system meets client expectations and is ready to be handed over to the client.

For this UAT test, we'd like to:

- Verify that the pipeline works as expected (upload file -> run model -> display statistics)
- Evaluate the accuracy of the AI action classification

**Out of Scope**

The following should not be used during testing:

- PCAP files captured from platforms other than YouTube (e.g. Netflix, general browsing) — the classification model has only been trained on YouTube traffic and will not produce meaningful action labels for other platforms
- PCAP files that do not contain any of the supported user actions (Like, Comment, Play, Search, Subscribe) — these will still display network metadata on the dashboard, but action classification results will not be meaningful and should not be used to evaluate model accuracy

# Testing team

In this section, members of the team along with their roles when undertaking user acceptance testing are listed.

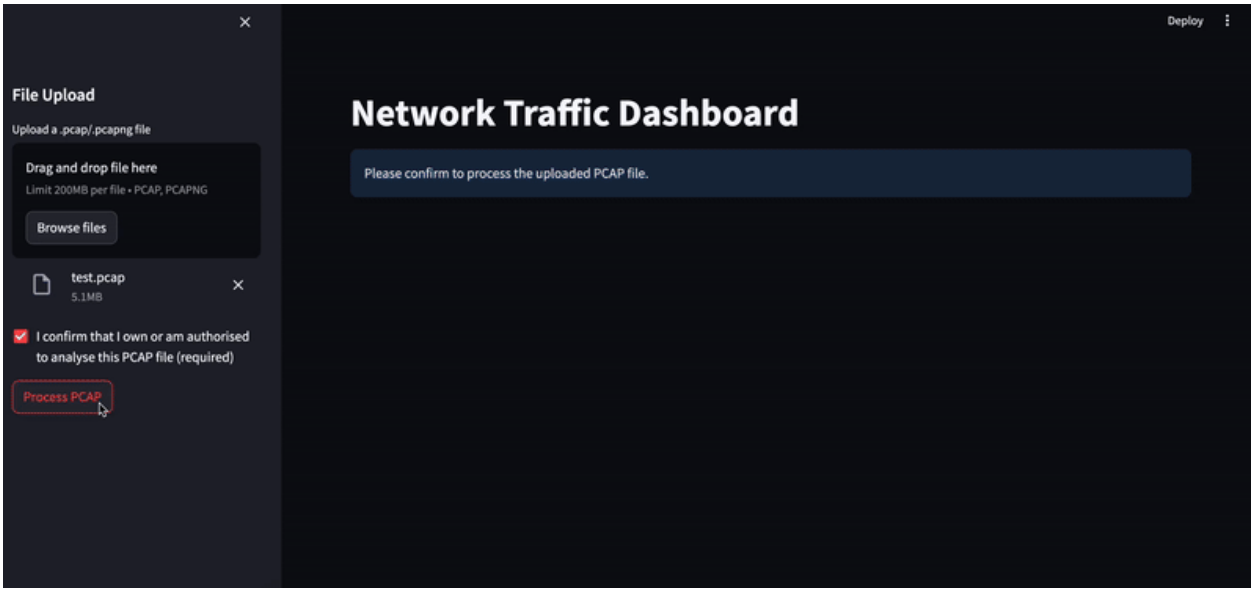
| **Name** | **Responsibilities** |
|---|---|
| Amelia | UAT Coordinator (handle communication between end users and QA team)<br>Design test cases<br>Consent form<br>UAT training |
| Sophia | Design test cases and UAT plan<br>Create test data and survey |
| Tim | Design test cases<br>Create test data<br>Linux testing environment setup |
| Tomiris | Design test cases<br>Create test data<br>Document results and observations |

# Milestones and deliverables

This section contains all deliverables for successful UAT execution.

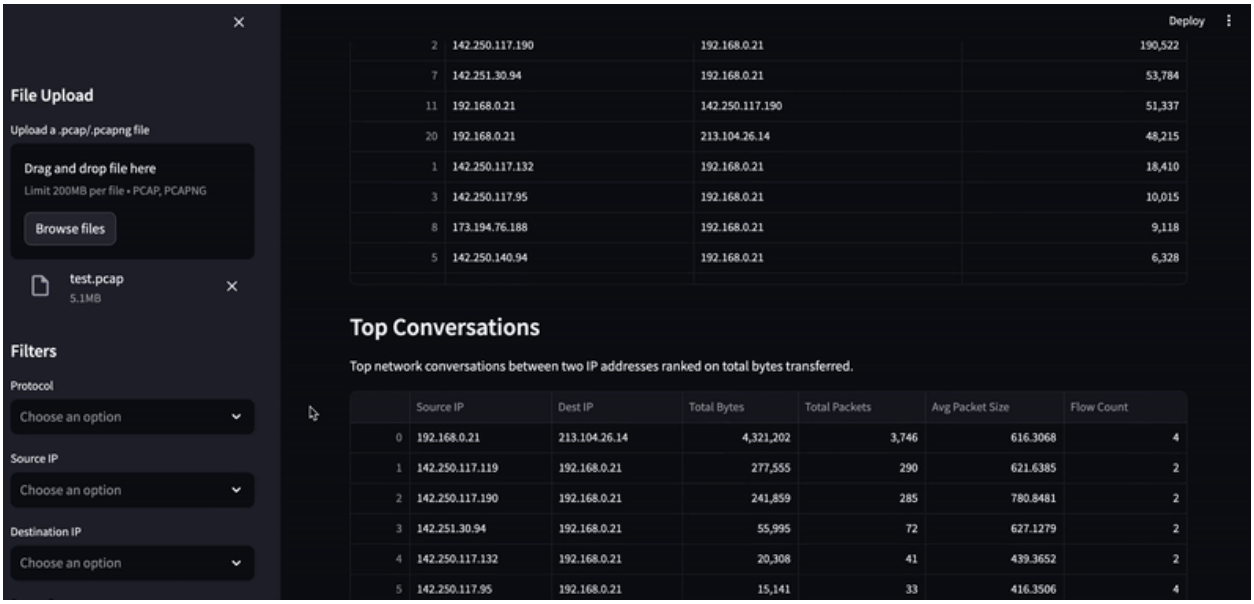
## Design & testing process overview (Tomiris)

Our system consists of a dashboard where, on the left panel, the user is prompted to upload their PCAP file. After the PCAP file is uploaded, it runs through our trained AI model in the background.




The user is then shown a set of diagrams on the main panel, displaying:

- A summary of the PCAP file activity and AI-classified user actions
- Metadata about the file, such as traffic volume, top endpoints, and conversations
- Segmented flows and their individual features




The testing will occur in 4 stages:

1. Staging environment: The staging environment will be set up by our team during the client meeting. This environment will closely replicate a real-world scenario in which the system is intended to operate. Testing will be carried out using our team members' personal devices, with results recorded accordingly. Packet capture (PCAP) files will be provided by our team to support the testing process.
2. Training: User Acceptance Testing (UAT) testers will be trained by Amelia. This training will include an overview of the UAT process, a walkthrough of the system's key features, and guidance on the type of feedback expected from testers.
3. UAT execution: Test cases have been developed collaboratively by all team members. During execution, Amelia will support testers in working through their assigned cases. Sophia and Tim will guide the client through the dashboard, while Tomiris will document results and observations.
4. Reporting: During the UAT, the results of all test cases will be recorded. In addition, user feedback on specific features will be collected through an online survey following the testing phase. Any issues identified will be addressed through system updates, after which a final round of UAT will be conducted.

# Staging environment (Sophia)

The staging environment will be set up with our team members' personal devices as follows:

Device 1: go through the test cases that are to be conducted with the client (Amelia)
Device 2: record test case results and observations during the testing (Tomiris)
Device 3: run pre-installed application, allow the client to use the dashboard and go through different PCAP file upload scenarios, including predictions from the dataset and manual recordings (Sophia)

Device 4: run pre-installed application and run main pipeline, running through the pipeline of extracting the dataset and training the model, go through dashboard tables and filtering (Tim)

The installed application is taken from our team's GitHub repository.
PCAP files used for the tests are pre-recorded using Wireshark/tcpdump and are saved on the corresponding personal devices.

# Training (Amelia)

The training phase ensures that the UAT participants understand how to use the system before executing test cases. Training will be delivered in a short, guided session prior to testing, either via Microsoft Teams or in-person.

**Training Content**
During the session, users will be introduced to the objectives of the UAT, the core system features, and what is expected of them during testing. A brief hands-on demonstration example will also be shown. During the training, any questions will be clarified to ensure users are well prepared.

By the end of the training session, users should be able to do the following:

- Upload and process a PCAP file independently
- Understand and interpret dashboard outputs
- Apply and clear filters correctly
- Download relevant data
- Complete assigned UAT test cases confidently

# UAT Execution (Tim)

Execution will take place over a single 60-minute session with the client. During this session, we will verify the processing pipeline from legal acknowledgment and PCAP upload, through interpretation of the activity timeline and anomaly prediction.

Steps:

1. **Onboarding:** Get the client set up with the system dashboard, demonstrate and explain the legal notice and explain what we expect of them.
2. **Test case execution:** The client will be given a specific test case with clear steps to take and made aware of any pre-requisites such as files required to complete a test.
3. **Debrief:** Once all test cases have been conducted, we will ask the client for brief feedback on the experience.
4. **Feedback:** The client will be sent a feedback form to complete after the testing process. Response data will be used to improve the system for further testing where necessary.

## Reporting & data analysis (Sophia)

Each test case will be recorded during the testing by the assigned member and reviewed against the expected results, identifying:

- Pass/Fail outcomes
- Observations (e.g. errors, feedback, confusion)

A Google Form survey will be sent out to the client after the UAT, covering:

- Usability
- Functionality
- Accuracy
- Encountered errors
- Open feedback
- Overall satisfaction

A second UAT will be held with an attempt to correct any potential errors.
The final findings will be summarized and included in the final client handover.

# Environmental requirements

## Hardware requirements (Tim)

Our system does not require special hardware to be run on and as such should run on most modern Windows, Linux and MacOS machines.

**Recommended Hardware:**
**RAM:** 8GB+ (Network traffic analysis typically requires RAM that is 2-5x the size of the raw PCAP file)
**CPU:** Quad-core or higher (Intel Core i5/Ryzen 5)
**Storage:** 10GB+ (Required to temporarily store uploaded files and dataset)
**Network:** No network connection is required to run locally but a network card is required

## Software requirements (Amelia):

The tester must have the following software and dependencies installed on their system in order to run the application successfully:

- Python 3.11
- Required Python libraries as listed in the ‘requirements.txt’ file
- PCAP test files (Provided by testers)
- A supported operating system (Windows, macOS, or Linux)

- A modern web browser (eg. Chrome, Edge, or Firefox) to access the dashboard interface

# Test cases

This section includes the features to be tested and will be referred to during the UAT.

## 1) ML Training

Test ID: MLT01
**Scenario:** Create dataset CSV

**Pre-requisites**
1. Dataset is set up correctly with a single directory for each 'action' type.
2. Open terminal with access to required python packages
3. Navigate to comp2003-2025-2026-team-15\SourceCode\network-traffic-profiler\src\ML\action_classification

**Test steps**
**1.** Run 'python generate_dataset.py'

**Pass/fail criteria**
- **Pass**: Observe new CSV created in '/../model_training' containing PCAP features
- **Fail**: CSV not created, corrupt or missing features

**Observations:** None
**Status:** Pass

---

Test ID: MLT02
**Scenario:** Train ML model

**Pre-requisites**
1. Feature set CSV was successfully generated in previous test (MLT01)
2. Open terminal with access to required python packages
3. Navigate to comp2003-2025-2026-team-15\SourceCode\network-traffic-profiler\src\ML\model_training

**Test steps**
**2.** Run 'python train_model.py'

**Pass/fail criteria**

- **Pass**: Observe model training complete and tests run successfully**,** new PKL files created
- **Fail**: Training fails, any test fails, or PKL files not created

**Observations:** Command return with errors ( unable to run)
**Status:** **Fail**

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## 2) Legal Notice & Compliance (Amelia)

**Test ID: LNC01**
**Scenario:** Legal notice is displayed before any file upload

**Pre-requisites**
- Application is opened
- No PCAP file has been uploaded

**Test steps**
1. Launch the application
2. Observe the main dashboard area and sidebar

**Pass/fail criteria**
- **Pass**:
  - Legal notice is clearly visible on screen
  - The notice is easily and clearly understood
  - Notice includes information about user responsibility and UK law

- **Fail**:
- Legal notice is missing or unclear/ not easily understood
- User can proceed without seeing legal guidance

**Observations:** None
**Status:** Pass

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**Test ID: LNC02**
**Scenario:** User cannot process file without confirming legal authorisation

**Pre-requisites**
- Application is opened
- Any valid PCAP file is selected/uploaded

**Test steps**

1. Upload a PCAP file
2. Do not tick the confirmation checkbox
3. Click “Process PCAP”

**Pass/fail criteria**

- **Pass**:
  - Error message appears:
    “You must confirm that you are authorised to analyse this PCAP file before proceeding.”
  - Processing does not begin

- **Fail**:
- File is processed without confirmation
- No error message is displayed

**Observations:** None
**Status:** Pass

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**Test ID: LNC03**
**Scenario: Full s**uccessful acknowledgement of legal compliance allowing processing

**Pre-requisites**
- Any valid PCAP file is uploaded

**Test steps**
3. Upload a PCAP file
4. Do tick the confirmation checkbox
4. Click “Process PCAP”

**Pass/fail criteria**

- **Pass**:
  - Processing begins successfully
  - No legal warning blocks progress

- **Fail**:
- Processing does not start despite confirmation

**Observations:** None
**Status:** Pass

# 3) File Upload & Pipeline

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**Test ID:** FUP01
**Scenario:** Large file warning

**Pre-requisites**
PCAP:
- FUP03.pcap

**Test steps**
1. Upload FUP03.pcap
2. Observe if large file size warning appears

**Pass/fail criteria**
- **Pass**: An warning appears 'Please note that large files may induce long processing times and require large amounts of RAM to be available on the host machine.'
- **Fail**: Warning does not appear

**Observations:** None
**Status:** Pass

---

**Test ID:** FUP04
**Scenario:** Successful authorisation and upload

**Pre-requisites**
PCAP:
- FUP02.pcap

**Test steps**
3. Upload FUP02.pcap
4. Confirm legal right to process
5. Press 'Process PCAP' button

**Pass/fail criteria**
- **Pass**:
  1. A box appears titled 'Processing'
  2. Progress indicators appear in-line with processing stage:
     a. 'Extracting network flows'
     b. 'Validating network flows'
     c. 'Predicting user actions'

d. ‘Detecting anomolies in the data’

- **Fail**: Any of the processing indicators do not appear

**Observations:** None
**Status:** Pass

---

**Test ID:** FUP05
**Scenario:** File swap logic

**Pre-requisites**
PCAP:
- FUP05.pcap

**Test steps**
6. Upload FUP05.pcap
7. Confirm legal right to process
8. Press ‘Process PCAP’ button

**Pass/fail criteria**
- **Pass**: Existing analysis data disappears and a new pipeline is triggered
- **Fail**: Existing analysis data is still visible, or processing indicators do not appear

**Observations:** None
**Status:** Pass

---

**Test ID:** FUP06
**Scenario:** Linux/Unix compatability

**Pre-requisites**
1. Ubuntu testing environment
2. PCAP:
   - FUP06.pcap

**Test steps**
1. Upload FUP06.pcap
2. Confirm legal right to process
3. Press ‘Process PCAP’ button

**Pass/fail criteria**

- **Pass**:
  1. A box appears titled 'Processing'
  2. Progress indicators appear in-line with processing stage:
     e. 'Extracting network flows'
     f. 'Validating network flows'
     g. 'Predicting user actions'
     h. 'Detecting anomolies in the data'
  3. Dashboard displays extracted results and ML classifications
- **Fail**: New pipeline is not created, pipeline does not complete as expected

**Observations:** None
**Status:** Pass

---

# 4) Activity Analysis (Action Classification)

**Test ID:** AA01
**Scenario:** PCAP data from the dataset (from model training) is uploaded to the dashboard

**Pre-requisites**

- PCAP files for each activity from the dataset (e.g. PCAP files/dataset/Comment01.pcap)

**Test steps**

1. Open /PCAP files/dataset folder
2. Upload PCAP file for each activity
3. Analyse results

**Pass/fail criteria**

- **Pass**: Accuracy mostly matches the expected predicted actions
- **Fail**: Accuracy doesn't match with the expected predicted actions

**Observations:** Comments : Rename 'Play' to 'Stream' for more accuracy as 'Play' would indicate the action of clicking the button
**Status:** Pass

---

**Test ID:** AA02
**Scenario:** User recorded PCAP data performing multiple and combined actions are uploaded to the dashboard

**Pre-requisites**
- Manually recorded PCAP files

**Test steps**
1. Open /testPCAPs/records folder
2. Upload PCAP files for combined activities
3. Analyse results

**Pass/fail criteria**
- **Pass**: Accuracy mostly matches the expected predicted actions
- **Fail**: Accuracy doesn't match with the expected predicted actions

**Observations:** None
**Status:** Pass

## 5) Traffic Filtering (Sidebar)

**Test ID:** TF01
**Scenario:** Filter flows by a single attribute (Protocol)

**Pre-requisites**
- Any valid PCAP file has been successfully uploaded and processed
- Dataset contains multiple protocol types (TCP/UDP)

**Test steps**
1. Locate the "Filters" section in the sidebar
2. Open the Protocol dropdown
3. Select one protocol
4. Observe all dashboard components

**Pass/fail criteria**
- **Pass**:
- Only flows matching the selected protocol are displayed
- All visualisations (pie chart, timeline, tables) update accordingly

- **Fail**: Data unrelated to the selected protocol is still visible (visualisations do not update)

**Observations:** Refine port number filtering for MacOs. Port numbers are being cast to a float so port 80 appears as 80.0 instead

**Status:** Pass

---

**Test ID:** TF02
**Scenario:** Apply multiple filters simultaneously

**Pre-requisites**
- Any valid PCAP file has been successfully uploaded and processed with multiple IPs and ports
- Dataset contains varied values across source/destination IPs and ports

**Test steps**
1. In the sidebar, select a Protocol, Source IP, Destination Port
2. Observe dashboard output

**Pass/fail criteria**
- **Pass**:
- Displays flows that satisfy all selected filters
- Charts and tables reflect the filtered dataset
- **Fail**:
- Filters behave independently instead
- Incorrect/extra/ unrelated data is still being displayed

**Observations:** None
**Status:** Pass

---

**Test ID:** TF03
**Scenario:** Clear filters and restore full dataset

**Pre-requisites**
- Any valid PCAP file has been successfully uploaded and processed
- Filters have already been applied
- Filtered dataset is visibly reduced correctly

**Test steps**
1. Remove all selected filter values
2. Observe dashboard output

**Pass/fail criteria**
- **Pass**:
- Full, unfiltered dataset is restored

- All charts and tables return to original state
- **Fail**:
- Data remains partially filtered / remain the same
- Dashboard does not refresh correctly

**Observations:** None
**Status:** Pass

## 6) Dashboard Tables (Tomiris)

**Test ID:** DT01
**Scenario:** All Flows table displays correct columns and description.

**Pre-requisites**

- .pcap files uploaded and processed, ‘All Flows’ checkbox enabled in Table Display Options

**Test steps**

1. Upload .pcap via the sidebar file uploader.
2. Confirm ownership and click 'Process PCAP'.
3. Scroll to the 'All Flows Table' section.
4. Read the description text above the table.
5. Inspect the column headers in the rendered table.

**Pass/fail criteria**

- **Pass**: All expected columns are present, correctly labelled, and the description text is accurate.
- **Fail**: One or more expected columns are missing, mislabelled, or the description is absent or incorrect.

**Observations:** None

**Status:** Pass

---

**Test ID:** DT02
**Scenario:** Traffic Volume Over Time graph renders and reflects data

**Pre-requisites**

- .pcap files uploaded and processed successfully

**Test steps**

1. Upload .pcap and process it.
2. Scroll to the 'Traffic Volume Over Time' section.
3. Verify the chart title and axis labels are visible.
4. Confirm bars are present along the x-axis.

**Pass/fail criteria**

- **Pass**: Bar chart renders with a visible x-axis (packet index) and y-axis (byte count), and bars are present for the uploaded traffic.
- **Fail**: Chart does not render, axes are missing or unlabelled, or no bars appear for a non-empty PCAP.

**Observations:** Graph is not really descriptive. The x and y-axis legend is not clearly/easily understood

**Status:** Pass

---

**Test ID:** DT03
**Scenario:** Top Endpoints table shows busiest IP pairs ranked by byte count

**Pre-requisites**

- .pcap uploaded and processed. 'Top Endpoints' checkbox enabled in Table Display Options.

**Test steps**

1. Upload mixed.pcap and process it.
2. Scroll to the 'Top Endpoints' section.
3. Read the description text above the table.
4. Inspect the columns: Source Address, Destination Address, Byte Count.
5. Verify the first row has the highest Byte Count value.

**Pass/fail criteria**

- **Pass**: Table renders with Source Address, Destination Address, and Byte Count columns. Rows are ordered from highest to lowest byte count.
- **Fail**: Table does not render, columns are missing, or rows are not sorted by byte count descending.

**Observations:** None

**Status:** Pass

## 7) Flagged Flows / Anomaly Detection Display (Tomiris)

**Test ID:** FFAD01
**Scenario:** Multicast destination flow appears in Flagged Flows with a reason

**Pre-requisites**

- multicast_dst.pcap is used and the dashboard is running

**Test steps**

1. Upload multicast_dst.pcap via the sidebar.
2. Confirm and process the file.
3. Scroll to the 'Flagged Flows' section.
4. Check that at least one row is present.
5. Read the 'Flagged Reason' column for the flagged flow.

**Pass/fail criteria**

- **Pass**: At least one row appears in Flagged Flows. The Flagged Reason column is non-empty and references the invalid destination or multicast address.
- **Fail**: No row appears in Flagged Flows, or the Flagged Reason column is blank or missing.

**Observations:** None

**Status:** Pass

---

**Test ID:** FFAD02
**Scenario:** Single-packet flow appears in Flagged Flows with low packet count reason

**Pre-requisites**

- single_packet.pcap is used and dashboard is running

**Test steps**

1. Upload single_packet.pcap via the sidebar.
2. Confirm ownership and click 'Process PCAP'.
3. Scroll to the 'Flagged Flows' section.

4. Locate the row corresponding to the single-packet flow.
5. Read the 'Flagged Reason' column.

**Pass/fail criteria**

- **Pass**: The single-packet flow appears in Flagged Flows and the Flagged Reason column states or references a low packet count.
- **Fail**: Flow does not appear in Flagged Flows, or Flagged Reason is blank or does not mention packet count.

**Observations:** None

**Status:** Pass

---

**Test ID:** FFAD03
**Scenario:** Anomalous Flows section shows empty state when no anomalies are detected

**Pre-requisites**

- no_anomalies.pcap with insufficient data is used

**Test steps**

1. Upload the no_anomalies.pcap file and process it.
2. Scroll to the Anomalous Flows section.
3. Observe whether a table or empty state message is shown.

**Pass/fail criteria**

- **Pass:** 'No anomalous flows detected' message is displayed. No empty table, error, or blank section is shown.
- **Fail:** An empty table renders, a Python traceback is shown, or the section is blank with no message.

**Observations:** None

**Status:** Pass

---

# 8) CSV download (Amelia)

**Test ID:** CSV01

**Scenario:** Download full dataset as CSV

**Pre-requisites**
- A PCAP file has been successfully uploaded and processed
- No filters are applied in the sidebar

**Test steps**
1. In the sidebar, click the “Download Filtered CSV” button
2. Open and observe the downloaded file

**Pass/fail criteria**
- **Pass**:
  - CSV file downloads successfully
  - File contains all flows from the dataset
  - Column headers match dashboard fields
- **Fail**:
  - File does not download
  - File is empty/corrupt

**Observations:** None
**Status:** Pass

---

**Test ID: CSV02**
**Scenario:** Download filtered dataset as CSV (protocol)

**Pre-requisites**
- Any valid PCAP file has been successfully uploaded and processed
- At least one filter applied (eg, Protocol)

**Test steps**
1. Apply one or more filters in the sidebar
2. Click the “Download Filtered CSV”
3. Open and observe the downloaded file
4. Compare contents with filtered dashboard

**Pass/fail criteria**
- **Pass**:
  - CSV file downloads successfully
  - File contains all flows from the selected filtered dataset
  - Column headers match dashboard fields
- **Fail**:
  - File does not download or empty/corrupt
  - CSV includes unfiltered data

- Mismatch between dashboard and CSV contents

**Observations:** None
**Status:** Pass

---

**Test ID: CSV03**
**Scenario:** Download CSV when filters result in no data

**Pre-requisites**
- A PCAP file has been successfully uploaded and processed
- Filters are applied in the sidebar to which there is zero matching flows

**Test steps**
1. In the sidebar, apply restrictive filters (eg. combination of IP and port) where there is no results
2. Observe the dashboard and try to download the CSV file

**Pass/fail criteria**
- **Pass**:
  - Option to download CSV file is not present
  - The dashboard shows no data is present
  - No system crash or error occurs
- **Fail**:
  - User is able to download a CSV file despite having no results

**Observations:** None
**Status:** Pass

---

# 9) Error Handling

**Test ID:** EH01
**Scenario:** An empty PCAP file or a PCAP file containing fake data is uploaded onto the dashboard

**Pre-requisites**
- Empty PCAP file (/PCAP files/errorHandling/empty.pcap)
- Fake PCAP file containing text (/PCAP files/errorHandling/fake.pcap)

**Test steps**
1. Open /PCAP files/errorHandling folder
2. Upload PCAP files to trigger error message

**Pass/fail criteria**

- **Pass**: Dashboard displays correct error message
- **Fail**: Dashboard crashes or shows backend error log trace

**Observations:** None
**Status:** Pass

---

**Test ID:** EH02
**Scenario:** A PCAP file containing no valid flows is uploaded onto the dashboard

**Pre-requisites**
- Valid PCAP file containing no valid flows (/testPCAPs/errorHandling/icmp_only.pcap)

**Test steps**
1. Open /testPCAPs/errorHandling folder
2. Upload PCAP file to trigger error

**Pass/fail criteria**
- **Pass**: Dashboard displays correct error message
- **Fail**: Dashboard crashes or shows backend error log trace

**Observations:** None
**Status:** Pass

---

**Test ID:** EH03
**Scenario:** Filters are applied on a valid PCAP file upload until no matching flows are found

**Pre-requisites**
- Any valid PCAP file

**Test steps**
1. Upload PCAP file
2. Select filters until no matched flows are found to trigger error message

**Pass/fail criteria**
- **Pass**: The dashboard displays a warning that no matching flows are found.
- **Fail**: The dashboard displays a trace log error and continues showing data.

**Observations:** None
**Status:** Pass

# Signoff

I hereby accept this final product:

_____

(Signature)

Name:
Date: